

DATA DICTIONARY & PRISMA SCHEMA SPECIFICATION

Analytic & Reporting Dashboard Divisi

TUJUAN

Menjadi sumber acuan struktur data untuk NestJS, Prisma ORM, PostgreSQL, API contract, dan automated test. Keputusan yang belum tersedia ditandai sebagai open decision, bukan diisi dengan asumsi tersembunyi.

Parameter	Baseline
Basis	PRD v0.4 + UI/UX v0.2 + ARD v0.3
Cakupan	Seluruh divisi; pilot Manager Minimarket
Database	PostgreSQL
Data access	Prisma ORM; Prisma Studio hanya untuk development
ID strategy	UUID untuk record utama (baseline)
Status	Ready untuk schema draft dan API contract

Cara membaca: mulai dari peta relasi, gunakan tabel field saat membuat schema.prisma, lalu gunakan constraint dan checklist saat menulis migration serta test.

1. Reading Guide & Data Ownership

Business rule	Model	Field	Constraint	Code
PRD / approval	Entity catalog	Dictionary	DB integrity	Prisma + test

1.1 Status Label

Label	Meaning	Vibe coder action
Required	Berasal dari kebutuhan yang disepakati	Implementasikan dan uji
Baseline	Keputusan teknis awal	Gunakan hingga ada pengganti
Open	Informasi bisnis belum tersedia	Jangan mengarang; buat configurable
Derived	Nilai dihitung dari data sumber	Hitung di server

1.2 Write Ownership

Domain	Writer	Reader
Identity & Organization	Superadmin / system	All modules
People & Documents	HRD	Manager, User, Reporting
Revenue	Manager / Superadmin	Dashboard, Targets, BOD
Targets	Manager; approval BOD	Dashboard, Reporting
Performance	Manager / Superadmin; approval BOD	BOD, Manager, User
Audit & Notification	System	Privileged users

CRITICAL RULE

Role memberi capability; data scope membatasi record. Kedua pemeriksaan wajib dilakukan di server.

2. PostgreSQL & Prisma Conventions

Data concept	Prisma / PostgreSQL baseline	Rule
Primary key	String @id @default(uuid()) @db.Uuid	Konsisten pada aggregate root
Timestamp	DateTime @db.Timestamptz(6)	Simpan UTC; konversi di UI
Business date	DateTime @db.Date	Tanpa jam
Money	Decimal @db.Decimal(18,2)	API memakai decimal string
Percentage	Decimal @db.Decimal(7,4)	Jangan gunakan Float untuk nilai final
Code / slug	String + @unique	Stabil walau nama berubah
Metadata	Json?	Hanya metadata/error
Soft delete	deletedAt DateTime?	Hanya master data yang perlu recovery

2.1 Naming

Layer	Convention	Example
Prisma model	PascalCase singular	RevenueDaily
Prisma field	camelCase	businessDate
PostgreSQL table	snake_case plural via @@map	revenue_daily
PostgreSQL column	snake_case via @map	business_date
Foreign key	<entity>Id	outletId
Enum value	UPPER_SNAKE_CASE	NEEDS_CONFIRMATION

OPEN DECISIONS

Konfirmasi sebelum production migration: format employee number, zona waktu tampilan, mata uang perusahaan, presisi uang, serta mapping status absensi. Baseline sementara: UTC storage dan IDR display.

3. Relationship & Entity Map

Division 1 - N Outlet	Outlet 1 - N Fact	Employee 1 - N Workforce	Period Target + score	Approval Locks version
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From	Relation	To	Integrity rule
Division	1 - N	Outlet	Outlet wajib memiliki division
Employee	1 - N	EmployeeAssignment	Simpan effective period
Outlet	1 - N	RevenueDaily	Satu row per business date
RevenueImport	1 - N	StagingRow	Source row immutable
RevenueTarget	1 - N	TargetApproval	Approval append-only
PerformanceTemplate	1 - N	TemplateIndicator	Total weight divalidasi
Assessment	1 - N	AssessmentScore	Satu score per indicator
FileObject	1 - N	EmployeeDocument	File private

3.1 Entity Catalog

Module	Models	Purpose
Identity	User, Role, Permission, UserRole, RolePermission	Authentication and authorization
Organization	Division, Outlet, UserScope	Hierarchy and access scope
People	Employee, EmployeeAssignment, Contract, Attendance, WorkSchedule, LeaveRequest	Workforce reporting
Files	FileObject, EmployeeDocument	Private employee files
Revenue	RevenueImport, StagingRow, RevenueDaily, RevenueMonthly, Reconciliation	Excel/POS canonical facts
Targets	RevenueTarget, TargetApproval	Proposal and BOD approval
Performance	KpiDefinition, Template, Indicator, Assessment, Score, Approval	Individual assessment
Control	WorkflowAction, AuditLog, Notification	Traceability

4. Common Fields & Historical Scope

Field	Type	Applies to	Rule
id	UUID	All primary models	Primary key
createdAt	Timestamptz	Mutable models	Default now
updatedAt	Timestamptz	Mutable models	@updatedAt
createdById	UUID?	Accountable writes	Actor user
status	Enum	Workflow models	Use transition method
statusChangedAt	Timestamptz?	Workflow models	Set with transition
version	Int	Versioned records	Starts at 1
deletedAt	Timestamptz?	Recoverable master	Never for audit/action

4.1 EmployeeAssignment

Field	Type	Req.	Rule
employeeId	UUID FK	Yes	Employee owner
divisionId	UUID FK	Yes	Historical division
outletId	UUID? FK	No	Null for division-level staff
effectiveFrom	Date	Yes	Start inclusive
effectiveTo	Date?	No	Null means current
createdById	UUID FK	Yes	HRD/system actor

No overlapping active assignments for one employee
 Transactions retain original divisionId/outletId snapshot
 Moving employee must not rewrite historical reports

5. Identity & Organization Dictionary

5.1 User, Division, Outlet

Model.Field	Type	Req.	Rule
User.email	String @unique	Yes	Normalized lowercase
User.passwordHash	String	Yes	Never returned by API
User.isActive	Boolean	Yes	Default true
User.employeeId	UUID? @unique	No	Employee account link
Division.code	String @unique	Yes	Division is data, not enum
Division.name	String	Yes	Display name
Outlet.divisionId	UUID FK	Yes	onDelete Restrict
Outlet.code	String @unique	Yes	Stable business code
Outlet.name	String	Yes	Display name
Outlet.isActive	Boolean	Yes	Import/scope validation

5.2 Roles & Scope

Model.Field	Type	Req.	Rule
Role.code	String @unique	Yes	BOD, SUPERADMIN, HRD, MANAGER, USER
Permission.code	String @unique	Yes	Capability code
UserRole.userId	UUID FK	Yes	Unique with roleId
UserRole.roleId	UUID FK	Yes	No implicit BOD approval
UserScope.divisionId	UUID? FK	No	Null only for explicit global scope
UserScope.outletId	UUID? FK	No	Outlet belongs to division
UserScope.effectiveFrom	Date	Yes	Historical scope
UserScope.effectiveTo	Date?	No	Null means active

6. People & Workforce Dictionary

6.1 Employee

Field	Type	Req.	Rule
employeeNo	String @unique	Yes	Format is open decision
fullName	String	Yes	Display/legal policy open
employmentStatus	EmploymentStatus	Yes	ACTIVE/INACTIVE/TERMINATED baseline
hireDate	Date	Yes	Cannot exceed endDate
endDate	Date?	No	Required when terminated
userId	UUID? @unique	No	Optional application account

6.2 Workforce Models

Model	Key fields	Mutation boundary
Contract	employeeId, type, startDate, endDate, status	HRD manages; others read only
Attendance	employeeId, attendanceDate, checkInAt?, checkOutAt?, status	Source open; dashboard read only
WorkSchedule	employeeId, workDate, shiftCode?, startAt, endAt	Source open; dashboard read only
LeaveRequest	employeeId, leaveType, startDate, endDate, status	HR source; dashboard read only

MVP BOUNDARY

Dashboard menyimpan data untuk pelaporan, tetapi bukan mesin absensi, penjadwalan, atau persetujuan cuti.

7. Employee Documents & Files

Model.Field	Type	Req.	Rule
FileObject.storageKey	String @unique	Yes	Private key, not public URL
FileObject.originalName	String	Yes	Sanitized display only
FileObject.mimeType	String	Yes	Allowlist
FileObject.sizeBytes	BigInt	Yes	Upload limit
FileObject.checksumSha256	String	Yes	Integrity support
EmployeeDocument.employeeId	UUID FK	Yes	Document owner
EmployeeDocument.documentType	DocumentType	Yes	Configurable later
EmployeeDocument.fileObjectId	UUID FK	Yes	Restrict while referenced
EmployeeDocument.version	Int	Yes	Unique per employee/type/version
EmployeeDocument.validUntil	Date?	No	Expiry alert

7.1 Access Log Minimum

```
actorUserId, employeeDocumentId, action
occurredAt, requestId, ipAddress?
result: ALLOWED | DENIED
short-lived signed URL after authorization
```

SECURITY

HRD menggunakan UI aplikasi untuk production. Prisma Studio bukan panel administrasi dokumen dan tidak boleh diekspos ke production user.

8. Revenue Import Dictionary

8.1 RevenueImport

Field	Type	Req.	Rule
importType	RevenueImportType	Yes	DAILY or MONTHLY
sourceType	RevenueSourceType	Yes	EXCEL or POS
fileObjectId	UUID? FK	Excel	Required for Excel
checksumSha256	String	Excel	Duplicate detection
periodMonth	Date/month key	Yes	Normalized period
status	ImportStatus	Yes	Workflow transition only
totalRows	Int	Yes	Default 0
validRows / invalidRows	Int	Yes	Sum equals totalRows
uploadedById	UUID FK	Yes	Manager/Superadmin
postedAt	Timestamptz?	No	After atomic post

8.2 RevenueStagingRow

Field	Type	Req.	Rule
revenueImportId	UUID FK	Yes	Import owner
rowNumber	Int	Yes	Unique per import
rawData	Json	Yes	Original parsed row
outletCode	String?	No	Mapped on validation
businessDate	Date?	Daily	Required for daily
grossRevenue / netRevenue	Decimal?	No	Validated >= 0
validationStatus	RowValidationStatus	Yes	PENDING/VALID/INVALID
errors	Json?	No	Field error codes

9. Revenue Facts & Reconciliation

9.1 Daily and Monthly Facts

Model.Field	Type	Req.	Rule
RevenueDaily.outletId	UUID FK	Yes	Unique with businessDate
RevenueDaily.businessDate	Date	Yes	Source of truth
RevenueDaily.grossRevenue	Decimal(18,2)	Yes	>= 0
RevenueDaily.netRevenue	Decimal(18,2)	Yes	>= 0
RevenueDaily.sourceImportId	UUID FK	Yes	Immutable lineage
RevenueMonthly.outletId	UUID FK	Yes	Unique with periodMonth
RevenueMonthly.periodMonth	Date/month key	Yes	Reconciliation input
RevenueMonthly.grossRevenue	Decimal(18,2)	Yes	Monthly recap
RevenueMonthly.netRevenue	Decimal(18,2)	Yes	Monthly recap
RevenueMonthly.sourceImportId	UUID FK	Yes	Immutable lineage

9.2 Reconciliation

Field	Type	Req.	Rule
outletId + periodMonth	UUID + month	Yes	One active version
dailyGross / dailyNet	Decimal	Yes	Derived sums
monthlyGross / monthlyNet	Decimal	Yes	Imported recap
grossDifference / netDifference	Decimal	Yes	Monthly minus daily
status	ReconciliationStatus	Yes	Workflow status
confirmedById	UUID? FK	No	Authorized reviewer
confirmationNote	String?	Conditional	Required for confirmed difference

TRUTH RULE

Monthly recap tidak pernah menimpa daily facts secara diam-diam. Selisih harus terlihat dan dikonfirmasi.

10. Revenue Target Dictionary

Model.Field	Type	Req.	Rule
RevenueTarget.outletId	UUID FK	Yes	Target per outlet
RevenueTarget.periodMonth	Date/month key	Yes	Target period
RevenueTarget.metricType	RevenueMetricType	Yes	GROSS or NET
RevenueTarget.amount	Decimal(18,2)	Yes	>= 0
RevenueTarget.version	Int	Yes	Version identity
RevenueTarget.status	ApprovalStatus	Yes	Workflow state
RevenueTarget.proposedById	UUID FK	Yes	Assigned Manager
RevenueTarget.submittedAt	Timestamptz?	No	Set on submit
TargetApproval.action	ApprovalAction	Yes	APPROVE or RETURN
TargetApproval.actorUserId	UUID FK	Yes	Must be BOD
TargetApproval.note	String?	Conditional	Required for RETURN
TargetApproval.occurredAt	Timestamptz	Yes	Append-only

10.1 Constraints

- Unique: outletId + periodMonth + metricType + version.
- Only one approved active target for outlet, period, and metric.
- Creator cannot approve the same target.
- Approved target is locked; reopen requires reason and audit.
- Division target is derived from approved outlet targets.

DO NOT DUPLICATE

Jangan simpan target divisi terpisah pada MVP jika nilainya hanya penjumlahan target outlet.

11. Performance Dictionary

11.1 KPI Template

Model.Field	Type	Req.	Rule
KpiDefinition.code	String @unique	Yes	Stable formula key
KpiDefinition.name	String	Yes	Display label
KpiDefinition.level	KpiLevel	Yes	INDIVIDUAL/OUTLET/DIVISION
KpiDefinition.unit	KpiUnit	Yes	SCORE/PERCENT/MONEY/COUNT
PerformanceTemplate.divisionId	UUID? FK	No	Null = reusable
PerformanceTemplate.version	Int	Yes	Immutable after use
TemplateIndicator.kpiDefinitionId	UUID FK	Yes	Indicator reference
TemplateIndicator.weight	Decimal(7,4)	Yes	Active total = 100%
TemplateIndicator.orderNo	Int	Yes	UI order

11.2 Assessment & Approval

Model.Field	Type	Req.	Rule
Assessment.employeeId	UUID FK	Yes	Individual KPI
Assessment.periodMonth	Date/month key	Yes	Assessment period
Assessment.templateId	UUID FK	Yes	Template version locked
Assessment.status	ApprovalStatus	Yes	Workflow state
Assessment.totalScore	Decimal?	Derived	Server calculated
AssessmentScore.indicatorId	UUID FK	Yes	Unique per assessment
AssessmentScore.rawScore	Decimal	Yes	Within range
AssessmentScore.weightedScore	Decimal	Derived	Server calculated
AssessmentApproval.action	ApprovalAction	Yes	BOD only
AssessmentApproval.note	String?	Conditional	Required for RETURN

KPI BOUNDARY

Omzet outlet tidak otomatis menjadi KPI individual. Aturan atribusi harus disetujui dan formula harus versioned.

12. Enum & Workflow Registry

Enum	Values (baseline)	Notes
EmploymentStatus	ACTIVE, INACTIVE, TERMINATED	Confirm other statuses
ContractStatus	ACTIVE, EXPIRED, TERMINATED	HRD managed
AttendanceStatus	PRESENT, LATE, ABSENT, LEAVE, OFF	Source mapping required
RevenueImportType	DAILY, MONTHLY	Excel granularity
RevenueSourceType	EXCEL, POS	POS reserved
ImportStatus	UPLOADED, VALIDATING, VALID, FAILED, POSTED, CANCELLED	No free update
RowValidationStatus	PENDING, VALID, INVALID	Staging only
ReconciliationStatus	OPEN, MATCHED, NEEDS_CONFIRMATION, CONFIRMED, LOCKED	Difference workflow
ApprovalStatus	DRAFT, SUBMITTED, APPROVED, RETURNED	Targets/assessments
ApprovalAction	APPROVE, RETURN, REOPEN	REOPEN privileged
RevenueMetricType	GROSS, NET	Target metric
KpiLevel	INDIVIDUAL, OUTLET, DIVISION	Reporting level

12.1 Transition Guard

DRAFT -> SUBMITTED: creator + validation pass
 SUBMITTED -> APPROVED: BOD + creator != approver
 SUBMITTED -> RETURNED: BOD + note required
 RETURNED -> DRAFT: authorized editor
 APPROVED -> REOPEN: privileged reason + audit

13. Constraints & Index Plan

Model	Constraint / index	Reason
Outlet	UNIQUE(code); INDEX(divisionId, isActive)	Lookup and filter
EmployeeAssignment	INDEX(employeeId, effectiveFrom, effectiveTo)	Historical scope
Attendance	UNIQUE(employeeId, attendanceDate)	No duplicate day
RevenueImport	UNIQUE(checksumSha256, importType, periodMonth)	Idempotency
StagingRow	UNIQUE(revenueImportId, rowNumber)	Source traceability
RevenueDaily	UNIQUE(outletId, businessDate); INDEX(businessDate)	Daily truth
RevenueMonthly	UNIQUE(outletId, periodMonth)	Monthly recap
Reconciliation	INDEX(outletId, periodMonth, status)	Action queue
RevenueTarget	UNIQUE(outletId, periodMonth, metricType, version)	Version identity
Assessment	UNIQUE(employeeId, periodMonth, templateId, version)	Version identity
AssessmentScore	UNIQUE(assessmentId, indicatorId)	One score
AuditLog	INDEX(entityType, entityId, occurredAt)	Trace query

13.1 Delete Behavior

Relation type	Behavior
Master referenced by history	Restrict delete; deactivate instead
Temporary staging before post	Cascade allowed in import cleanup
Audit / approval action	Never cascade; append-only
File referenced by document	Restrict until authorized removal

14. Prisma Schema Starter

SCOPE

Contoh menunjukkan pola, bukan schema lengkap. Terapkan audit fields, @@map, relation, dan onDelete secara konsisten.

```
datasource db { provider = "postgresql"; url = env("DATABASE_URL") }
generator client { provider = "prisma-client-js" }

model Division {
  id String @id @default(uuid()) @db.Uuid
  code String @unique @db.VarChar(50)
  name String @db.VarChar(150)
  isActive Boolean @default(true) @map("is_active")
  outlets Outlet[]
  createdAt DateTime @default(now()) @db.Timestamptz(6) @map("created_at")
  updatedAt DateTime @updatedAt @db.Timestamptz(6) @map("updated_at")
  @@map("divisions")
}

model Outlet {
  id String @id @default(uuid()) @db.Uuid
  divisionId String @db.Uuid @map("division_id")
  code String @unique @db.VarChar(50)
  name String @db.VarChar(150)
  division Division @relation(fields: [divisionId], references: [id], onDelete: Restrict)
  @@index([divisionId])
  @@map("outlets")
}
```

15. Prisma Pattern: Revenue Fact

```
model RevenueDaily {
  id String @id @default(uuid()) @db.Uuid
  outletId String @db.Uuid @map("outlet_id")
  businessDate DateTime @db.Date @map("business_date")
  grossRevenue Decimal @db.Decimal(18, 2) @map("gross_revenue")
  netRevenue Decimal @db.Decimal(18, 2) @map("net_revenue")
  sourceImportId String @db.Uuid @map("source_import_id")
  outlet Outlet @relation(fields: [outletId], references: [id], onDelete: Restrict)
  createdAt DateTime @default(now()) @db.Timestamptz(6) @map("created_at")
  @@unique([outletId, businessDate])
  @@index([businessDate])
  @@map("revenue_daily")
}
```

15.1 Service Rules

- Gunakan Prisma transaction untuk post import, submit, approve, return, dan reopen.
- Policy dan scope diperiksa di server sebelum query/write.
- Serialize Prisma Decimal sebagai string ke API.
- Raw SQL hanya untuk kebutuhan terukur dan tetap parameterized.
- Prisma Studio dibatasi untuk local/development.

VALIDATION LAYERS

DTO validation memeriksa bentuk input; domain service memeriksa aturan bisnis; PostgreSQL constraint menjaga integritas terakhir.

16. Migration, Seed & Handoff

16.1 Migration Order

Order	Migration group	Exit check
1	Identity + Organization	Roles, divisions, outlets load
2	People + assignments	No overlapping demo assignment
3	Files + documents	Private reference enforced
4	Revenue + reconciliation	Daily uniqueness and lineage pass
5	Targets + approvals	BOD-only transition tests pass
6	Performance + approvals	Weight and locking tests pass
7	Audit + notification	Trace query indexed

16.2 Seed Policy

- Development seed: one user per role, six divisions, sample Minimarket outlets, and anonymous demo data.
- Production seed: permission, role, and minimum config only; no default password or employee example.
- Seed must be idempotent and environment-safe.

16.3 Vibe Coding Prompt

PROMPT STARTER

Buat atau perbarui [MODEL/MIGRATION] mengikuti Data Dictionary v0.2 dan ARD v0.3. Gunakan PostgreSQL mapping, UUID, Decimal untuk uang, onDelete eksplisit, index/unique constraint, transaction, serta unit/integration test. Tandai field atau enum baru sebagai asumsi.

16.4 Ready for API Contract

- Model dan field utama disetujui.
- Open decisions diselesaikan atau dibuat configurable.
- Prisma schema dapat validate dan migration berjalan pada database kosong.
- Development seed serta role/scope tests berhasil.
- Data sensitif dan delete behavior telah direview.

17. SobatHR Canonical Data Dictionary

Keputusan: Model kanonik dipisahkan dari format ekspor SobatHR. Nama kolom vendor disimpan pada mapping/import layer, bukan dijadikan struktur domain utama.

17.1 Integration & Import Models

Model	Field minimum	Constraint / purpose
HrIntegrationSource	id, provider, transport, active, configVersion	Satu provider dapat berganti FILE ke API tanpa mengubah facts.
HrImportBatch	id, sourceId, domain, periodStart, periodEnd, checksum, status, counts, uploadedBy, timestamps	Unique candidate: sourceId + domain + period + checksum.
HrImportRow	id, batchId, rowNum, externalEmployeeId, normalizedPayload, validationStatus, errors	Staging only; payload payroll perlu masking/encryption policy.
ExternalEmployeeMap	id, sourceId, externalEmployeeId, employeeId, validFrom, validTo	Unique active mapping per source + externalEmployeeId.

17.2 Canonical HR Facts

Model	Grain	Field inti
AttendanceFact	Employee + workDate	employeeId, outletId, scheduledIn/Out, actualIn/Out, status, lateMinutes, batchId
LeaveFact	Employee + request/source ID	employeeId, type, startDate, endDate, duration, status, batchId
PermissionFact	Employee + request/source ID	employeeId, date/time range, type, duration, status, batchId
OvertimeFact	Employee + workDate/request ID	employeeId, approvedMinutes, actualMinutes, status, batchId
SicknessFact	Employee + request/source ID	employeeId, startDate, endDate, duration, status, batchId; tanpa diagnosis
PayrollSummaryMonthly	Month + company/division	periodMonth, divisionId nullable, employeeCount, grossTotal, deductionTotal, netTotal, currency, batchId

17.3 Enum Registry

Enum	Values awal	Status
HrTransport	FILE, API	Confirmed design; API availability open.
HrDomain	ATTENDANCE, PERMISSION, LEAVE, OVERTIME, SICKNESS, PAYROLL	Confirmed scope.
HrImportStatus	UPLOADED, VALIDATING, READY, PUBLISHED, PARTIAL, FAILED, SUPERSEDED	Confirmed workflow baseline.
HrRowStatus	VALID, INVALID, UNMAPPED	Confirmed baseline.
DataCompleteness	COMPLETE, PARTIAL, MISSING	Confirmed UI/API contract.
AttendanceStatus	Vendor values mapped later	Open until SobatHR export is obtained.

17.4 Constraints & Privacy

- Facts wajib membawa sourceId/importBatchId agar setiap angka dapat ditelusuri ke asalnya.
- Unmapped employee rows tidak boleh dipublikasikan ke canonical facts.
- PayrollSummaryMonthly tidak memiliki employeeId; detail payroll tidak masuk query model dashboard.
- Nilai kosong dan nilai nol harus dibedakan. Missing/partial status wajib disimpan bersama agregasi.
- Soft correction menggunakan superseded batch; audit history tidak boleh dihapus oleh impor berikutnya.
- Retensi staging/raw payload, money precision, currency, dan definisi gross/deduction/net membutuhkan keputusan final.

17.5 Prisma Implementation Note

Guardrail: Buat model import lebih dahulu, kemudian mapping employee, canonical facts, dan terakhir query summaries. Jangan membuat tabel berdasarkan setiap nama sheet vendor.

- Gunakan Decimal untuk nilai uang dan DateTime untuk timestamp; tanggal kerja dapat dinormalisasi pada UTC boundary dengan display timezone perusahaan.
- Tambahkan composite unique/index setelah contoh ekspor nyata memvalidasi grain dan external identifiers.
- Pisahkan payroll module dan endpoint agar policy BOD-only dapat diuji secara mandiri.